

MultiCloudApp AWS CloudOps Platform

Overview

MultiCloudApp is a **CloudOps Command Center** built on AWS EKS. It demonstrates containerized deployments, monitoring, and AI anomaly detection adaptable for any industry (Healthcare, BFSI, Travel, Retail).

Current Achievements

- AWS EKS cluster provisioned with Terraform
- Dockerized frontend & backend pushed to AWS ECR
- Kubernetes deployments & services running
- NodePort/LoadBalancer services exposed
- PowerShell transcripts for reproducibility

Future Goals

- Multicloud expansion (Azure, GCP)
- Mobile app layer (React Native / Flutter)
- Sectorspecific dashboards (Hospital, BFSI, Travel, Retail)
- CI/CD pipelines with GitHub Actions
- AIpowered anomaly detection integrated with Prometheus

Quick Start

```
“powershell terraform init terraform apply aws eks update-kubeconfig --region ap-south-1 --name multicloud-eks kubectl apply -f k8s/
```

2. Supporting Markdown Files (in /docs)

- **SETUP.md** Installation steps for Windows (Terraform, Docker, kubectl, Helm, AWS CLI).
- **MONITORING.md** Prometheus & Grafana setup, dashboard screenshots.
- **AI.md** IsolationForest anomaly detection script + integration notes.
- **FUTURE.md** Roadmap for hospital, BFSI, travel, retail use cases.

Documentation Links

Explore detailed guides inside the /docs folder: - Setup Guide - Monitoring Guide - AI Anomaly Detection - Architecture Overview - Future Roadmap # Setup Guide

Prerequisites

- Windows 10/11
- PowerShell
- Docker Desktop
- AWS CLI
- Terraform
- kubectl
- Helm

Steps

1. **Provision EKS Cluster** “powershell terraform init terraform apply aws eks update-kubeconfig –region ap-south-1 –name multicloud-eks
2. Build & Push Images docker build -t multicloud-frontend . docker tag multicloud-frontend .dkr.ecr.ap-south-1.amazonaws.com/multicloud-frontend:v3 docker push .dkr.ecr.ap-south-1.amazonaws.com/multicloud-frontend:v3
3. Deploy to Kubernetes kubectl apply -f k8s/frontend-deployment.yaml kubectl apply -f k8s/frontend-service.yaml kubectl apply -f k8s/backend-deployment.yaml kubectl apply -f k8s/backend-service.yaml
4. Verify

kubectl get pods kubectl get svc

MONITORING.md

“markdown # Monitoring Setup

Prometheus

“powershell helm repo add prometheus-community https://prometheus-community.github.io/helm-charts helm install prometheus prometheus-community/prometheus

Grafana

helm repo add grafana https://grafana.github.io/helm-charts helm install grafana grafana/grafana

Dashboards CPU usage

Memory usage

Pod health

API latency

Access Grafana via the LoadBalancer external IP.

AI.md

“markdown # AI Anomaly Detection

Script Example

```
“python import pandas as pd from sklearn.ensemble import IsolationForest
data = pd.DataFrame({"cpu":[20,22,19,95,21,20]}) model = IsolationFor-
est(contamination=0.1) model.fit(data) print("Anomaly detection results:",
model.predict(data))
```

Integration Connect to Prometheus metrics

Detect anomalies in CPU/memory

Trigger alerts or autoscaling

FUTURE.md

“markdown # Future Roadmap

MultiCloud Expansion

- Azure AKS
- GCP GKE
- Hybrid deployments

Mobile Applications

- **Android (React Native / Kotlin)**
- **iOS (React Native / Swift)**
- Consume backend APIs
- Provide dashboards for different sectors

Sector Use Cases

- Hospital Patient monitoring dashboards
- BFSI Transaction anomaly detection
- Travel Booking and cluster health visualization
- Retail Inventory and sales monitoring

CI/CD

- GitHub Actions
- Automated builds and deployments # AI Anomaly Detection

Overview

This module adds intelligence to the CloudOps platform by detecting unusual behavior in system metrics (CPU, memory, latency). It helps prevent outages and supports predictive scaling.

Example Script

```
“python import pandas as pd from sklearn.ensemble import IsolationForest
```

Sample CPU usage data

```
data = pd.DataFrame({"cpu": [20, 22, 19, 95, 21, 20]})
```

Train anomaly detection model

```
model = IsolationForest(contamination=0.1) model.fit(data)
```

Predict anomalies (-1 = anomaly, 1 = normal)

```
print("Anomaly detection results:", model.predict(data)) # Architecture
```

Current Achievements

- AWS ECR Docker images
- AWS EKS Kubernetes cluster
- Terraform Infrastructure provisioning
- Services NodePort/LoadBalancer
- Scripts Secret rotation, AI anomaly detection

Diagram

AWS Cloud

ECR (Docker Registry) multicloud-frontend:v3 multicloud-backend:latest

EKS Cluster Frontend Deployment Backend Deployment Services (LoadBalancer)

Monitoring Prometheus Grafana

Scripts Secret Rotation Cert Rotation AI Anomaly Detection

Future Expansion

- Multicloud (Azure, GCP)
 - Mobile apps (Android/iOS)
 - Sectorspecific dashboards
-

FUTURE.md

“markdown # Future Roadmap

MultiCloud Expansion

- Extend deployments to **Azure AKS** and **GCP GKE**.
- Hybrid cloud support for resilience and cost optimization.
- Unified Terraform modules for all providers.

Mobile Applications

- **Android (React Native / Kotlin)** and **iOS (React Native / Swift)**.
- Consume backend APIs for dashboards and alerts.
- Provide native mobile experience for operators and endusers.

Sector Use Cases

- **Hospital** Patient monitoring dashboards, appointment tracking.
- **BFSI** Transaction anomaly detection, fraud alerts.
- **Travel** Booking status dashboards, cluster health visualization.
- **Retail** Inventory and sales monitoring, demand forecasting.

CI/CD

- GitHub Actions pipelines for automated builds and deployments.
- Continuous testing and security scans.
- Multienvironment deployments (dev, staging, prod).

Security Automation

- Automated **secret rotation**.
- Certificate renewal scripts.
- ECR cleanup and image lifecycle policies.

Resume Impact

This roadmap demonstrates the ability to: - Build a **multicloud, fullstack platform**. - Extend into **mobile applications** for crosssector use. - Integrate

AI and monitoring for enterprise reliability. === Terraform State === No state.

```
=== EKS Cluster === { "cluster": { "name": "multiCloudAppCluster",
"arn": "arn:aws:eks:ap-south-1:654654230083:cluster/multiCloudAppCluster",
"createdAt": "2026-09-26T21:25:45.515000+05:30", "version": "1.36",
"endpoint": "https://247E3718C7712F06B1E00D1D53854807.gr7.ap-south-
1.eks.amazonaws.com", "roleArn": "arn:aws:iam::654654230083:role/EKSRole",
"resourcesVpcConfig": { "subnetIds": [ "subnet-01cc93764b916f596", "subnet-
0bc3bb43d487ef5e5" ], "securityGroupIds": [ "sg-0a6fa3bad81f7b634"
], "clusterSecurityGroupId": "sg-0a1e37df2d8541e59", "vpcId": "vpc-
027bdb9741258fbc7", "endpointPublicAccess": true, "endpointPrivateAccess":
false, "publicAccessCidrs": [ "0.0.0.0/0" ], "controlPlaneEgressMode":
"AWS_MANAGED" }, "kubernetesNetworkConfig": { "serviceIpv4Cidr":
"10.100.0.0/16", "ipFamily": "ipv4", "elasticLoadBalancing": { "en-
abled": false } }, "logging": { "clusterLogging": [ { "types": [ "api",
"audit", "authenticator", "controllerManager", "scheduler" ], "enabled":
false } ] }, "identity": { "oidc": { "issuer": "https://oidc.eks.ap-south-
1.amazonaws.com/id/247E3718C7712F06B1E00D1D53854807" } }, "status":
"ACTIVE", "certificateAuthority": { "data": "LS0tLS1CRUdJTiBDRVJUSUZJQ0FUR
S0tLS0tCk1JSURNakND...0FURS0tLS0tCk1JSURNakND...",
"active": { "id": "c6a5ae7a-2567-336c-8f61-bb239d7c2daf", "activatedBy":
"EKS" } }, "platformVersion": "eks.14", "tags": {}, "accessConfig": {
"authenticationMode": "CONFIG_MAP" }, "upgradePolicy": { "supportType":
"EXTENDED" }, "computeConfig": { "enabled": false, "nodePools": [] }, "stor-
ageConfig": { "blockStorage": { "enabled": false } }, "deletionProtection": false,
"controlPlaneScalingConfig": { "tier": "standard" }, "kubeApiServerConfig": {
"eventTtl": "60m", "serviceNodePortRange": { "minPort": 30000, "maxPort":
32767 } }, "kubeSchedulerConfig": { "nodeResourcesFit": { "scoringStrategy":
{ "type": "LeastAllocated", "resources": [ { "name": "cpu", "weight": 1 }, {
"name": "memory", "weight": 1 } ] } } }, "kubeControllerManagerConfig": {
"podGcControllerConfig": { "terminatedPodGcThreshold": 12500 }, "horizon-
talPodAutoscalerControllerConfig": { "horizontalPodAutoscalerSyncPeriod":
"15s" } } } }
```

```
=== Nodes === NAME STATUS ROLES AGE VERSION INTERNAL-IP
EXTERNAL-IP OS-IMAGE KERNEL-VERSION CONTAINER-RUNTIME
ip-172-31-22-72.ap-south-1.compute.internal Ready 3h v1.36.4-eks-f4fc4f1
172.31.22.72 13.233.75.222 Amazon Linux 2023.12.20260918 6.18.48-
109.150.amzn2023.x86_64 (amd64) containerd://2.2.7+unknown
```

```
=== Pods === NAMESPACE NAME READY STATUS RESTARTS
AGE IP NODE NOMINATED NODE READINESS GATES default
backend-567c657ff-kv7l4 1/1 Running 0 129m 172.31.18.62 ip-172-31-22-72.ap-
south-1.compute.internal default frontend-6d558c869c-rm4fz 1/1 Running
0 92m 172.31.16.21 ip-172-31-22-72.ap-south-1.compute.internal default
nginx-7f8fbb96d-8rm6t 1/1 Running 0 3h32m 172.31.19.91 ip-172-31-22-
72.ap-south-1.compute.internal kube-system aws-node-ggrzq 2/2 Running
```

```
0 3h 172.31.22.72 ip-172-31-22-72.ap-south-1.compute.internal kube-system
coredns-645b58c966-6w66t 1/1 Running 0 3h8m 172.31.17.74 ip-172-31-22-72.ap-
south-1.compute.internal kube-system coredns-645b58c966-dtf6b 1/1 Running 0
3h3m 172.31.20.85 ip-172-31-22-72.ap-south-1.compute.internal kube-system
kube-proxy-p9fsw 1/1 Running 0 3h 172.31.22.72 ip-172-31-22-72.ap-south-
1.compute.internal
```

```
=== Services === NAMESPACE NAME TYPE CLUSTER-IP EXTERNAL-
IP PORT(S) AGE SELECTOR default backend-service NodePort 10.100.141.170
5000:32551/TCP 129m app=backend default frontend-service NodePort
10.100.59.146 3000:32552/TCP 125m app=frontend default kubernetes
ClusterIP 10.100.0.1 443/TCP 3h44m default nginx NodePort 10.100.66.9
80:32550/TCP 176m app=nginx kube-system eks-extension-metrics-api
ClusterIP 10.100.42.232 443/TCP 3h44m kube-system kube-dns ClusterIP
10.100.0.10 53/UDP,53/TCP,9153/TCP 3h43m k8s-app=kube-dns
```

```
=== Deployments === NAMESPACE NAME READY UP-TO-DATE
AVAILABLE AGE CONTAINERS IMAGES SELECTOR default backend 1/1 1
1 129m backend 654654230083.dkr.ecr.ap-south-1.amazonaws.com/multicloud-
backend:v1 app=backend default frontend 1/1 1 1 110m frontend
654654230083.dkr.ecr.ap-south-1.amazonaws.com/multicloud-frontend:v3
app=frontend default nginx 1/1 1 1 3h32m nginx nginx app=nginx
kube-system coredns 2/2 2 2 3h43m coredns 602401143452.dkr.ecr.ap-south-
1.amazonaws.com/eks/coredns:v1.14.3-eksbuild.16 eks.amazonaws.com/component=coredns,k8s-
app=kube-dns
```

```
=== Docker Images === REPOSITORY TAG IMAGE ID CRE-
ATED SIZE 123456789012.dkr.ecr.ap-south-1.amazonaws.com/multicloud-
frontend v3 2b8fff5525c6 2 hours ago 709MB 654654230083.dkr.ecr.ap-
south-1.amazonaws.com/multicloud-frontend v3 2b8fff5525c6 2 hours
ago 709MB multicloud-frontend latest 2b8fff5525c6 2 hours ago 709MB
654654230083.dkr.ecr.ap-south-1.amazonaws.com/multicloud-frontend v1
0ef1c2d9457c 6 hours ago 709MB multicloudapp-frontend latest 623c7be321ff
6 hours ago 709MB multicloudapp-backend latest 91ec98d4e4c4 7 hours ago
133MB 654654230083.dkr.ecr.ap-south-1.amazonaws.com/multicloud-backend v1
f8217a76582b 7 hours ago 133MB multicloud-backend latest f8217a76582b 7
hours ago 133MB 123456789012.dkr.ecr.ap-south-1.amazonaws.com/multicloud-
backend latest f8217a76582b 7 hours ago 133MB thavrix-infra-thavrix-api
latest 8bd9fa850f55 3 months ago 148MB fcb8664233c1 3 months ago 142MB
prom/prometheus latest d2f7aaa363e1 4 months ago 436MB thavrix-infra-
thavrix-web latest f31ae8e802c5 4 months ago 62.5MB 918a2d9c996a 4 months
ago 62.5MB f671c4de62c5 4 months ago 142MB node 24-slim f5706f589c48
4 months ago 226MB postgres 15-alpine 5cce759a2777 4 months ago 274MB
grafana/grafana latest ffe38074db41 4 months ago 1.07GB hello-world latest
e2ac70e7319a 6 months ago 10.1kB
```

```
=== ECR Repositories === { "repositories": [ { "repositoryArn":
"arn:aws:ecr:ap-south-1:654654230083:repository/multicloud-frontend", "reg-
```

```
istryId": "654654230083", "repositoryName": "multicloud-frontend", "repositoryUri": "654654230083.dkr.ecr.ap-south-1.amazonaws.com/multicloud-frontend", "createdAt": "2026-09-26T23:19:45.972000+05:30", "imageTagMutability": "MUTABLE", "imageScanningConfiguration": { "scanOnPush": false }, "encryptionConfiguration": { "encryptionType": "AES256" } }, { "repositoryArn": "arn:aws:ecr:ap-south-1:654654230083:repository/multicloud-backend", "registryId": "654654230083", "repositoryName": "multicloud-backend", "repositoryUri": "654654230083.dkr.ecr.ap-south-1.amazonaws.com/multicloud-backend", "createdAt": "2026-09-26T22:58:06.869000+05:30", "imageTagMutability": "MUTABLE", "imageScanningConfiguration": { "scanOnPush": false }, "encryptionConfiguration": { "encryptionType": "AES256" } } ] }
```